

Use-Case Flow: Confirm Ride Booking

Peer-to-Peer Carpooling & Expense Splitter - Problem Statement #26

Primary Actor

Rider Commuter

Secondary Actors

Driver Host, Payment Gateway (external system)

Preconditions

- The Rider Commuter and Driver Host both have verified accounts (license and vehicle documents verified per FR-004).
- The Rider has previously requested a route match and received at least one candidate match within the 1.5 km detour tolerance (FR-001).
- The Driver Host's ride offer still has at least one available seat.

Postconditions

- Success: A seat is reserved on the Driver Host's trip, the fare split is calculated and recorded, payment is authorized, and both parties receive a confirmation notification.
- Failure: No booking is created and the Rider is notified of the reason (match invalid or payment declined).

Main Success Scenario

1. Rider Commuter selects a proposed match from the list of available Driver Host routes.
2. System re-validates that the selected match is still within the 1.5 km detour tolerance.
3. Rider Commuter taps "Confirm Booking" to request the seat.
4. System invokes the «include» Calculate Fare Split use case, computing the Rider's cost share from trip distance, current fuel price, and total number of co-riders.
5. System displays the calculated fare share to the Rider for review.
6. Rider Commuter approves the fare share and authorizes payment.
7. System connects to the Payment Gateway and the payment is authorized.
8. System reserves the seat, decrements the Driver Host's available seat count, and records the fare split.
9. System sends a confirmation notification with trip details and fare share to both the Rider Commuter and the Driver Host.
10. Use case ends successfully.

Alternate Flow - Detour No Longer Within Tolerance

- System detects the Driver Host's route has changed and the detour to the Rider's pickup point now exceeds 1.5 km.
- System displays a "Match no longer valid" message and removes the match from the Rider's list.
- System suggests the next-best available match within tolerance, if one exists.
- Rider Commuter selects an alternate match (flow returns to step 1) or cancels the request, ending the use case.

Alternate Flow - Payment Declined

- System displays a "payment failed" message.
- Rider Commuter is prompted to try another payment method.
- If payment fails again after 2 attempts, the booking is cancelled, the seat is released back to the Driver Host, and the Rider is notified.